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Using a depth camera for crop row detection and mapping for under-canopy navigation of agricultural robotic vehicle

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ABSTRACT

Computer vision provides local environmental information for robotic navigation in crop fields. It is particularly useful for robots operating under canopies of tall plants such as corns (*Zea Mays*) and sorghums (*Sorghum bicolor*), where GPS signal is not always receivable. However, the development of under-canopy navigation systems is still an open research area. The key contribution of our work is the development of a vision-based system for under-canopy navigation using a Time-of-Flight (ToF) camera. In the system, a novel algorithm was used to detect parallel crop rows from depth images taken under crop canopies. Two critical tasks in navigation were accomplished based on the detection results: 1) generating crop field maps as occupancy grids when reliable robot localization is available (from other sources such as GPS and IMU), and 2) providing inter-row vehicle positioning data when the field map is available and the localization is not reliable. The proposed system was evaluated in field tests. The test results showed that the proposed system was able to map the crop rows with mean absolute errors (MAE) of 3.4 cm and 3.6 cm in corn and sorghum fields, respectively. It provides lateral positioning data with MAE of 5.0 cm and 4.2 cm for positioning in corn and sorghum crop rows, respectively. The potential and limitations of using ToF cameras for under-canopy navigation were discussed.

1. Introduction

Robotic techniques were incorporated into agricultural production and research to automate various field operations (Bergerman et al., 2016). The robotic techniques increase the operational efficiency and reduce the labor cost when compared with traditional methods. Many agricultural field operations were already automated through automating heavy-duty field equipment, such as seeding (Pedersen et al., 2017), weeding (Steward et al., 2019) and harvesting (Birrell et al., 2019). However, lighter-duty and complicated field operations such as plant inspection and phenotyping usually require reduced human interaction while maintaining higher precision and robustness, imposing challenges to the intelligence and autonomy levels of the robot (Mueller-Sim et al., 2017). An important technology which determines the level of autonomy is autonomous navigation. Navigation system design remains as a challenge for a field robot, due to the complex and unstructured/semi-structured outdoor environment compared with the indoor conditions (Bergerman et al., 2016). It is particularly true for a small robot working under canopies of crop plants.

Many commercialized field robots employ Global Navigation Satellite Systems (GNSS) data as a global positioning source, and operate with a field map surveyed through aerial imaging or seed maps generated by planters (Bonadies and Gadsden, 2019). But the performance of these navigation systems highly depends on the GNSS signal quality. Especially for light-weight autonomous robots operating in corn and sorghum fields, the GNSS signal might be blocked by the canopies and the robots need to detect crop rows and avoid potential collisions on-the-go.

Computer vision-based techniques have been explored in recent years for navigation in crop fields. Since the crop rows present straight, parallel, and regularly spaced line patterns, the main strategy for vision-based navigation in crop field is to detect the crop rows using vision sensors. The row detection results were used for both robot localization and field mapping: with pre-generated field maps, the inter-row position of the robot was calculated and acts as an additional localization source for row guidance. Inversely, with reliable localization results, the crop row detection results were used to generate or update the field map for future navigation.

Most reported studies use above-canopy color cameras for crop row

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detection with which multiple rows are observed simultaneously (English et al., 2014; Li et al., 2019; Li et al., 2020; Morio et al., 2017). These algorithms rely on the reflectance difference between crop plants and the background. Therefore, challenges of these algorithms lie in the presence of weeds, and the connected inter-row plant canopies in above-canopy images, which usually happen at late growth stages. In addition, such algorithms are not suitable for robots navigating under tall plant canopies in narrow rows, where the sensing environment is cluttered due to the weeds and dropping leaves, and there is only limited crop plant visibility from the two adjacent rows (Fig. 1).

Another series of algorithms used data acquired from sensors placed under-canopy, which were mainly used by small robots operating under canopies of tall row crops such as corn and sorghum, or robots operating between tree rows in orchards (Shalal et al., 2013). When using color cameras, the pixels of plants, soil, and sky were segmented in the images, and the boundaries were extracted for directrix estimation. Yang et al. (2018) proposed an algorithm to detect the centerlines of maize rows under canopy by extracting the bottom section of plants. However, the nature of inconsistent illumination and shadow in the crop field would pose a challenge to these color image-based algorithms.

Since 3D features are less prone to the illumination inconsistency, sensors which are capable of directly supplying distance measurements, such as light-detection-and-ranging sensor (LiDAR) and ToF camera, have advantages compared to other cameras which measure reflectance intensity. LiDAR was broadly used in various of navigational applications including SLAM (Zhang and Singh, 2014) and autonomous driving (Wang et al., 2017). Interests of applying LiDAR in agriculture was sparked due to the reduced sensor price in recent years. In case of row-following tasks, the use of single-layer LiDAR sensors was reported in orchard navigation (Bergerman et al., 2015; Jones et al., 2019) and in less structured crop fields (Higuti et al., 2019; Mueller-Sim et al., 2017). In these applications, the general strategy was to place a single-layer LiDAR sensor levelly and close to the ground, aiming at the tree trunks or the crop stalks. The directrix was therefore measured by applying line fitting algorithms such as least square, RANSAC and PEARL to the measured data points (Isack and Boykov, 2012).

According to Higuti et al. (2019), there are several challenges posed to single-layer LiDAR-based navigation in the field. One is the cluttered rows where weeds and leaves may block the tree trunks or the crop plant stalks in the sensor output. Another is the loss of detailed 3D data since only a 2D set of distances measurement is available. In addition, frequent sensor obstruction can be caused by weeds and leaves, especially when navigating between crop rows. These challenges can be alleviated by using a sensor with a wider vertical view angle. A rotating 2D LiDAR system was developed to output 3D data for tree trunk detection in orchards, and the performance was better than using a single 2D LiDAR sensor (Zhang et al., 2013). Similarly, a 3D LiDAR-based navigation system was reported to outperform single-layer LiDAR-based systems during row following in pergola structured orchards (Bell et al., 2016).

Time-of-flight camera is another range sensor which measures distance based on the time of flight of modulated IR light (Li, 2014). It

produces depth images of a scene with low computational overheads at high acquisition rates. It has a wider vertical view angle compared to LiDAR sensors. ToF cameras were widely used in indoor navigation and mapping (Behrje et al., 2018; Weingarten et al., 2004; Yuan et al., 2009). But to our knowledge, there is no published research or commercial product on using ToF cameras as the primary sensor source for in-field, under-canopy navigation and mapping. That was because most commercially available ToF sensors come with weak light sources, therefore have poor measurement accuracy under directly sunlight, because of spectrum conflict between the sensor's active light source and the sunlight. Since some recently available ToF sensors have strong light sources, which enable the sensor to function in outdoor environments, the potential of using ToF cameras for under-canopy navigation becomes worthwhile to investigate.

This reported research was set out to investigate the feasibility of using a front-facing Time-of-Flight (ToF) camera to detect crop rows for navigation under the canopy of crops with narrow row spacing, such as corn and sorghum. The specific objectives were to (1) assess the developed under-canopy ToF-camera-based crop row detection algorithm; (2) evaluate the mapping and inter-row vehicle localization performance using the proposed row-detection algorithm.

This paper is organized as follows: In section 2, our developed robot PhenoBot 3.0 as the experimental platform is introduced, and the pipeline of crop row detection, inter-row vehicle localization and map generation is described. Section 3 provides the results of quantitative analyses of the localization and mapping. The capability and limitations of the proposed method and the ToF-based system are discussed in depth. Finally, the study is concluded in section 4.

2. Materials and methods

2.1. PhenoBot 3.0

This section describes a field-based ground robot, PhenoBot 3.0, as part of the project. PhenoBot 3.0 is a robotic ground vehicle designed for high-quality organ-level phenotypic data acquisition for corn and sorghum plants within an agronomic row spacing of 0.76 m (30") (Fig. 2, left). The robot was equipped with a telescoping (2–3.5 m) and auto-balancing sensor mast. A series of stereo cameras were equipped to acquire close-range side-view images of the plants (Fig. 2, middle). The telescoping mast and the motorized camera placement enable the robot to image the plants at different heights simultaneously. Various organ-level traits can be acquired by imaging including brace roots, stalks, ears, leaf angles, and tassels or panicles of maize and sorghum plants at different growth stages. The auto-balancing feature of the mast prevents the mast from colliding with the crop plants while the robot is running on uneven field surfaces, and keeps the cameras at a relatively constant imaging distances to the plants to ensure image quality. Our team has also developed a stereo camera module with an active illumination, which further increases the image quality under variable outdoor lighting conditions.

The robot rover features a narrow body design of 0.51 m (20") width

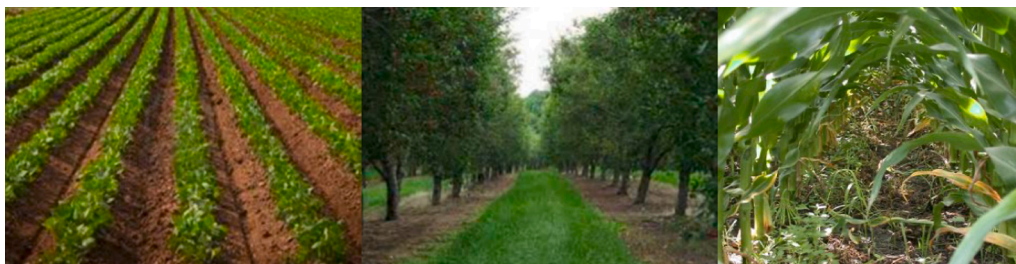


Fig. 1. Images for field-based navigation. Left: images of crop rows acquired above-canopy in field; Middle: images of tree rows acquired between rows in orchard; Right: images of crop rows acquired under tall plant canopies in field, where the crop rows are more difficult to detect.



Fig. 2. PhenoBot 3.0, an autonomous robotic ground vehicle designed for field-based corn and sorghum plant phenotyping within a 30-inch row spacing.

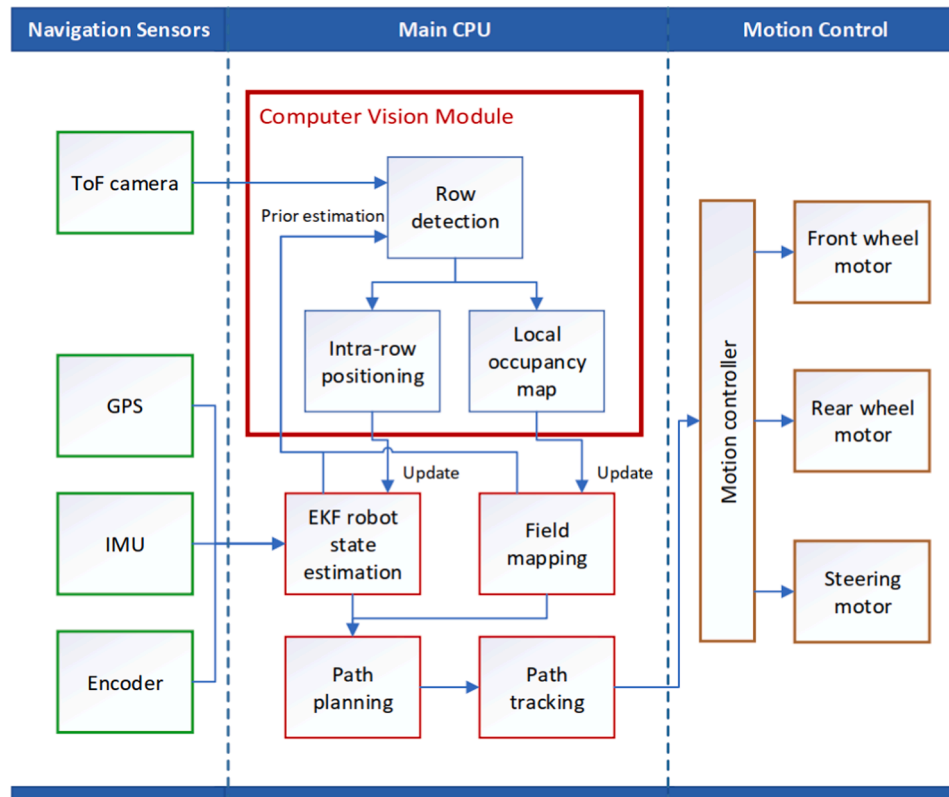


Fig. 3. The architecture of the PhenoBot 3.0 navigation system. (inter-row vehicle localization).

to fit into the required 0.76 m (30") row spacing. The rover body consists of the front and the rear part, connected by an articulated steering joint. In each body part, one DC brushed motor controls the rotation of two wheels through a differential gearbox. The central articulated steering mechanism and the differential gear-based power transmission enable the robot to swiftly traverse between narrow crop rows with high energy efficiency (Fig. 2, right). A set of high-capacity batteries were placed inside the robot, which can support up to 8 h driving in the field – enough time to survey about 1.8 ha (4.85 acres) at 3.2 Km/h (2 mph)

travel speed.

2.2. Navigation system architecture and sensors

The navigation system of PhenoBot 3.0 features a centralized control scheme (Fig. 2). It comprises a main computer running the control software, a series of sensors and motion actuators. The central CPU running Robotic Operation System (ROS) (Stanford Artificial Intelligence Laboratory et al, 2018) middleware handles the sensor data and

executes control algorithms, then sends commands to low-level motor controllers to control the motion of the robot. The navigation software consists of a computer vision module which detect the crop rows for estimating the lateral position and heading of the robot between crop rows, and creating a grid map of the crop rows. The rest are non-vision modules for robot pose estimation, mapping, path planning and tracking. The computer vision module is the focus of this reported research.

In PhenoBot 3.0, various sensors were used to accomplish the navigation task, including a ToF camera, an RTK-GPS module, two Inertia measurement units (IMU), and a set of encoders. The main sensor for vision-based navigation functions is a StarForm Swift-G ToF camera (Odos Imaging, Scotland). The camera equips with seven strong infrared LED light sources, which enables the camera to operate in outdoor conditions. It has view angles of $43^\circ \times 33^\circ$, and outputs depth and infrared intensity images simultaneously with a resolution of 640×480 . The precision of depth measurement is in centimeter-level when operating at the range of 0.3 m–2.5 m, which is sufficient for crop row detection. The camera was mounted 0.23 m (8") above the ground on the front part of the rover with a forward facing pitch angle of 8° downward from the horizontal direction (Fig. 4).

The other sensors act as additional localization sources for the robot pose estimation. An RTK GPS (Reach RS2, Emlid, Russia) antenna was mounted on the top of the sensor mast, which provides centimeter level global positioning of the robot. The mast tilt angle was tracked by an IMU (Adafruit BNO055, Adafruit, NY, USA) on the base of the mast to correct the deviation of GPS due to the tilting mast. Another IMU (3DM-GX5-15, LORD Sensing MicroStrain, USA) was placed in the front part of the rover, which measures the heading change rate and acceleration of the robot. Encoders were placed at the output shafts of the drive motors and the steering joint, to provide the speed and the steering angle of the robot. These signals were fused by an Extended Kalman Filter (Moore and Stouch, 2016) to calculate the position and heading of the robot in the world coordinate.

In this paper, the computer vision module was focused on. The rest of the modules in the navigation system will be described in our other studies. In our developed vision-based navigation module, there are

three major functions (Fig. 5). The first is a novel image processing algorithm for the detection of parallel crop rows from ToF depth and intensity images. The second is field mapping, which uses the row detection results and a robot pose estimation algorithm to generate crop row maps as occupancy grids. The last function is the estimation of the robot position and heading relative to the crop rows. In our application, the desired level of error of field mapping and lateral positioning is less than 5 cm (2") to avoid collision with the crop rows.

2.3. Crop row detection

The basic strategy of the row detection is to fit two parallel planes in the point cloud with a prior distance estimation using vegetation pixels representing the crop rows. In the row detection function, the depth image and the infrared intensity image from the depth sensor were taken as inputs. The framework of the crop row detection and localization algorithm is outlined as the following steps:

Step 1: Preprocessing: This procedure filtered the depth image using a modified bilateral filter and removed invalid pixels in the depth image.

Step 2: Ground detection: The filtered depth image was transformed into a 3D point cloud. The soil surface was detected (assumed to be a plane) in the 3D point cloud. Vegetation points above the soil surface were extracted.

Step 3: Crop row detection: The vegetation points were sliced and clustered into supervoxels to increase the percentage of representative points of crop rows (i.e. the corn stalk pixels). A linear programming method was applied to fit the parallel crop rows.

Corn and sorghum plants have similar appearance in their vegetative stage (Assefa et al., 2014). Therefore, the proposed row detection algorithm is applicable to both corn and sorghum. The algorithm was tuned and tested for corn and sorghum plants at different growth stages with different weed infestation levels. Each of these steps will be further described in the following sub-sections.

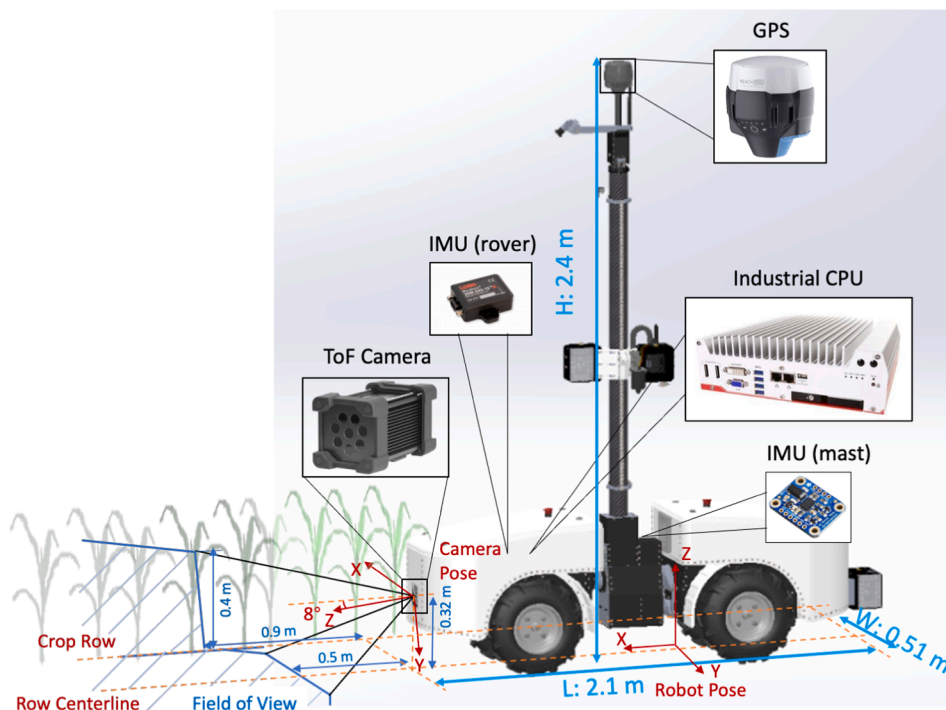


Fig. 4. The placement of the navigation sensors on PhenoBot 3.0, and the field of view of the ToF camera between crop rows.

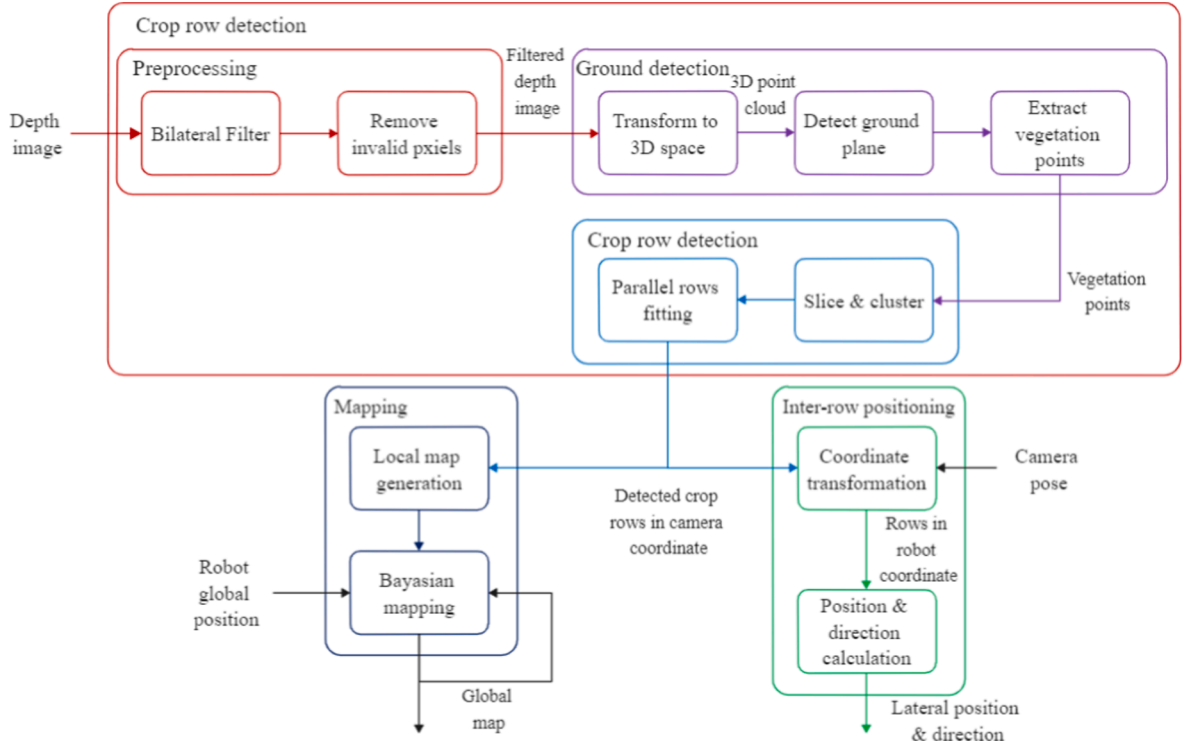


Fig. 5. Image processing pipeline of the developed vision-based navigation module.

2.3.1. Preprocessing

The raw depth images acquired outdoor by the ToF sensor contained a substantial amount of noise and flying pixels that need to be filtered. However, the infrared intensity image was observed less noisy (Fig. 6). Therefore, a modified bilateral filter which uses both depth image and infrared intensity image was developed to reduce the noise level in depth image. Then a pass-through filter was applied to remove the invalid points.

Bilateral filter is an edge-preserving smoothing algorithm (Paris et al., 2007), and it is widely used for improving the quality of the depth image. The original algorithm acts as a weighted averaging of pixels in a search window, during which both the difference of depth value of the pixels and the pixel distance on the image plane were taken into consideration. The depth value D of each pixel p after filtering can be denoted as $BF[D]_p$ by:

$$BF[D]_p = \frac{1}{W_p} \sum_{q \in S} G_{\sigma_s}(\|p - q\|) \cdot G_{\sigma_r}(D_p - D_q) \cdot D_q \quad (1)$$

where q is pixel in the search window S centered at p . G_{σ_s} is a spatial Gaussian term that decreases the influence of pixels further away from p , and G_{σ_r} is a range Gaussian term that decreases the influence of pixels with more different depth values than that of p . W_p is a normalization factor, which is the sum of weights in the search window. Both the spatial Gaussian and the range Gaussian term are calculated from the same image.

In this application, in order to take the advantage from the intensity image, the bilateral filtering was modified to calculate the range Gaussian term G_{σ_r} from the infrared intensity image I . Therefore, the depth value D of each pixel p in the depth image after filtering becomes:

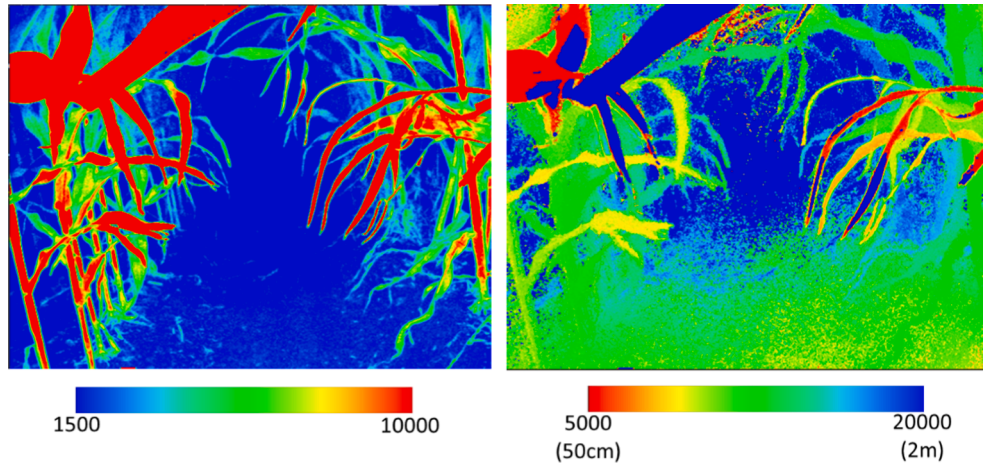


Fig. 6. Example raw IR intensity image (left) and depth image (right) from the ToF camera. Images were taken in a corn field (40 DAP). The color represents the raw intensity/depth value of each pixels. The depth measurement contains a substantial level of noise.

$$BF[D]_p = \frac{1}{W_p} \sum_{q \in S} G_{\sigma_s}(\|p - q\|) \cdot G_{\sigma_r}(I_p - I_q) \cdot D_q \quad (2)$$

Additionally, a passthrough filter was applied to reject pixels with lower measurement accuracy, which are the pixels either too far or too close to the camera. In this application, the pixels with depth less than 0.3 m or greater than 2.5 m were removed, based on the specifications of the camera.

2.3.2. Point cloud generation and ground detection

After filtering the depth image, a 3D point cloud was back-projected from the depth image using the camera projection matrix:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \cdot d \quad (3)$$

where (x, y) are the pixel coordinates in the depth image after rectifying the lens distortion, d is the depth value of the pixel, (f_x, f_y) are the focal length along the camera X and Y axis, respectively, (c_x, c_y) are the principle points along the camera X and Y axis, respectively, (X, Y, Z) are the 3D coordinates of the depth pixel. The parameters of projection and distortion were obtained through camera calibration (Zhang, 2000).

The ground plane was detected in the point cloud by fitting a 3D plane with Random Sample Consensus (RANSAC) (Fischler and Bolles, 1981)). Due to the presence of weeds, the detection result had a chance of returning incorrect results. Thus, the detection result was verified with the prior estimation of the camera pose to improve the detection robustness. If the detection result is consistent with the estimation, the detection result was kept. Otherwise, it was discarded, and the prior estimation was used for future steps.

The points above the ground were extracted into a new point cloud, which was composed of vegetation pixels from the crop rows and potentially weeds. The detected ground was used to serve as a reference plane for row detection since the crop plants were assumed to be perpendicular to the ground.

2.3.3. Row detection

As observed in the raw images, the pixels of corn/sorghum stalks served as a structural feature in crop row detection. But the stalk pixels were difficult to extract directly due to the presence of substantial amount of leaf pixels in the extracted point cloud and the noisy data even after filtering. In this application, the point cloud was clustered into supervoxels using an clustering algorithm based on horizontally slicing (Bao et al., 2019; Xiang et al., 2019) to reduce the proportion of leaf pixels in the point cloud. The clustering algorithm was originally developed to skeletonize the side-view point cloud of maize plants, and was proved to be effective in emphasizing the stalk features of tall growing plants such as maize and sorghum. In the clustering algorithm, the point cloud was first evenly sliced into thin layers along the growth direction of the plants. Afterwards, a Euclidean Clustering algorithm (Rusu and Cousins, 2011) was applied to cluster each layer into supervoxels based on the spatial connectivity. And the 3D centroid point of each supervoxel was extracted into a new point cloud (Fig. 7).

By assuming the crop rows are straight and parallel, the row detection problem was abstracted as a mathematical problem: fitting a pair of parallel planes (n_1, n_2) with prior estimated distance s_0 in 3D point clouds. A dynamic programming-based algorithm was applied to solve the problem. A pair of parallel planes in 3D space can be defined by four parameters: $(\theta, \varphi, s, \rho)$. The polar angle θ and the azimuthal angle φ indicate the direction of the normal vector of the plane in spherical coordinates. s is the distance between the parallel planes, and ρ is the distance from the origin to the central plane of the parallel planes toward the normal direction. (n_1, n_2) are four-element homogeneous vectors representing planes in 3D coordinate, and their equations can be expressed by:

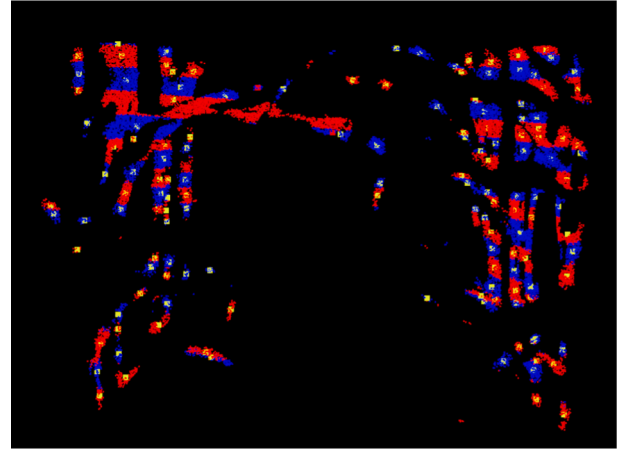


Fig. 7. The slicing-based clustering algorithm to enhance the structural features of corn plants. The adjacent slices are differentiated by red/blue colors. The extracted centroids of individual clusters are color in yellow. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$$n_1 \cdot (x, y, z, 1) = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta, \rho - \frac{s}{2}) \cdot (x, y, z, 1) = 0 \quad (4)$$

$$n_2 \cdot (x, y, z, 1) = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta, \rho + \frac{s}{2}) \cdot (x, y, z, 1) = 0 \quad (5)$$

A cost function to minimize was defined as:

$$cost(\theta, \varphi, s, \rho) = w_1 \sum_{p \in N_1} \frac{dis^2(p, n_1)}{size(N_1)} + w_2 \sum_{p \in N_2} \frac{dis^2(p, n_2)}{size(N_2)} + \gamma(s - s_0)^2 \quad (6)$$

in which N_1 and N_2 are the disjoint unions of the points in the point cloud which are closer to n_1 and n_2 , respectively. $size(N_1)$ and $size(N_2)$ are the number of points in N_1 and N_2 , respectively. s_0 is the prior estimation of the distance between the planes. The cost function is composed of three terms. The first two terms are the weighted average squared distances to the nearest plane. The weight factors w_1 and w_2 are defined based on the size of N_1 and N_2 as:

$$w_1 = \frac{\sqrt{size(N_1)}}{\sqrt{size(N_1)} + \sqrt{size(N_2)}} \quad (7)$$

$$w_2 = \frac{\sqrt{size(N_2)}}{\sqrt{size(N_1)} + \sqrt{size(N_2)}} \quad (8)$$

which sets a higher weight to the plane consists of more inliers, since more inliers usually means higher fidelity. The third term is the violation of the distance prior estimation, and its weight is controlled by a weight factor γ . The cost function is non-linear due to the conditional operation to dynamically determine the plane inliers.

The Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm (Chambers and Fletcher, 2001), which is an iterative method for solving unconstrained nonlinear optimization problems, was applied to minimize the cost function. In the algorithm, the approximated the Hessian matrix of the cost function was calculated approximately using local gradients during each iteration, and the parameters gradually approach the optimal value to minimize the cost function. Therefore, the optimal parameters $(\theta, \varphi, s, \rho)$ representing the estimated position and direction of the crop rows can be calculated. In addition, if the crop rows are assumed vertical to the ground, the problem can be reduced by fixing parameter θ to 0.

2.4. Mapping

In navigation applications, grid maps are used to represent the occupancy status of different locations in the world. In grid maps, the space is discretized into cells to approximate the obstacles in the real world, and each grid stores the probability that the cell is occupied by an obstacle. Given the crop row detection results, a local 2D grid map representing the crop row locations was constructed in real-time. The grid map was pie-shaped, with a radius of 2.5 m, which is consistent with the sensor's field of view (Fig. 8a). The cell size was set to 2.5 cm based on the concerns of navigation accuracy requirement, the sensor accuracy, and the computational cost. Based on the beam model of range finders theory (Thrun et al., 2005), the cells in the local map were divided into three categories: free space, occupied space, and unknown space, with occupancy probability of 0.3, 0.7 and 0.5, respectively. The categorization is performed according to the detected crop row position.

Based on the available global pose of the robot, the local map was used to update the global occupancy map (Fig. 8b). Using the Bayesian Mapping strategy (Thrun et al., 2005) the probability of occupancy of each cell was updated based on the Bayesian rule. This strategy is popularly applied in various mapping applications with LiDAR sensors, and the algorithm was implemented based on the ROS package "local_map". After the construction of the probability map, a global map of collision-free space was extracted by thresholding the occupancy probability, which provided references for collision-free navigation to the robot.

2.5. Robot inter-row localization

The direction and the lateral position of the robot relative to the crop row were estimated using the crop row detection results (n_1, n_2), which are four-element homogeneous vectors representing the detected left and right crop row planes in the camera coordinate, defined in Eqs. (4) and (5). The detected crop rows in robot coordinate $n_{R,i}$ ($i = 1, 2$) can be calculated:

$$n_{R,i} = \left((T_R^C)^{-1} \right)^T * n_i (i = 1, 2) \quad (9)$$

in which T_R^C is the 4×4 transformation matrix which transforms point in camera coordinate into robot coordinate. Therefore, the lateral position ρ_R and direction φ_R relative to the crop row centerline can be calculated:

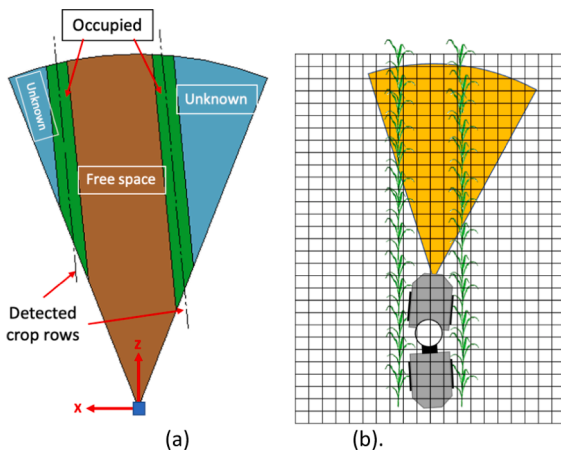


Fig. 8. The map and the mapping area. (a) The circular sector shaped local occupancy map with three occupancy states determined by the detected crop rows. (b) The updated area (in yellow color) in the global map, which is determined by the camera pose. The grid size is for representation and does not match the actual size. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$$\rho_R = (n_{R,1}(4) + n_{R,2}(4)) / 2 \quad (10)$$

$$\varphi_R = \arctan(n_{R,1}(1) / n_{R,1}(2)) \quad (11)$$

in which $n_{R,1}(i), n_{R,2}(i)$ are the i th element in the vectors representing the left and right crop rows, respectively. The results were further filtered using a Kalman filter, in which the system was modeled based on the lateral position and direction error model for path tracking (Karkee and Steward, 2010):

$$\begin{pmatrix} \rho_R \\ \varphi_R \end{pmatrix}_n = \begin{pmatrix} 1 & v \cdot \Delta t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \rho_R \\ \varphi_R \end{pmatrix}_{n-1} + \begin{pmatrix} 0 \\ \Delta t \end{pmatrix} \omega_{n-1} \quad (12)$$

in which ω is the angular speed of the robot, measured from the EKF-based robot state estimation module (Fig. 3), used as input of the filter. v is the forward speed of the robot. Δt is the time interval between two measurements, and the measurements are the measured lateral position and direction error (ρ_R, φ_R) using proposed algorithm.

3. Results and discussions

3.1. Crop row detection performance evaluation

The row detection algorithm was evaluated in terms of the accuracy of the estimated ground plane and row planes parameters using corn images. Images for evaluating the developed crop row detection algorithm were collected in the summer of 2018 using our developed PhenoBot 3.0 robot as described in the material section. The images were taken at two different times at two different locations in order to test the performance of the algorithm under different conditions such as different plant growth stages, weed infestation levels, and soil conditions:

The first dataset was collected in a corn breeding farm located in Omaha, Nebraska. The corn plants were at about the V5 growth stage, 30 days after planting (30 DAP). The soil was dry. The second dataset was collected at Iowa State University Agricultural Engineering and Agronomy Research Farm located in Boone, Iowa. The plants were at about the R1 growth stage, 80 DAP. The soil was wet. The fields all featured a standard 0.76 m (30") row spacing, and about 0.15 m (6") inter-plant spacing. About 3000 images were collected each time. The weather condition was sunny during data collection, and the sunlight illuminance ranged from 60,000 to 80,000 lux, and the camera exposure parameters were adjusted to fit the illumination conditions.

In this study, 120 images from the 30 DAP dataset and 200 images from the 80 DAP dataset were randomly sampled and manually labeled. More images were sampled from the 80 DAP dataset because the detection task is more challenging due to its higher weed infestation level, more complicated crop plant structure, and more frequent sensor blockage by the leaves. To quantify the performance of the algorithm, the accuracy of two critical steps: ground detection and crop row detection, were evaluated.

Several parameters were calculated from the detected ground and crop row plants for performance evaluation. In the ground detection step, two parameters were calculated from the detected ground plane, including the normal direction and the distance to the centroid of the point cloud (D2C) (Fig. 9a). In the crop row detection step, three parameters were calculated for each plane, including the normal direction, the distance from the crop rows to the camera origin position (D2O), and to the cloud centroid (D2C) (Fig. 9b). Therefore, accuracies of the ground detection and crop row detection were evaluated by comparing the above parameters of the detected planes with the ground truth. The ground truth was generated by manually labeling the ground plane and the crop row planes in the point cloud based on human observation, and calculating the corresponding parameters using the labeled planes. The annotation was done by one person for a uniform standard. Each image was annotation three times and the average was taken. The cloud

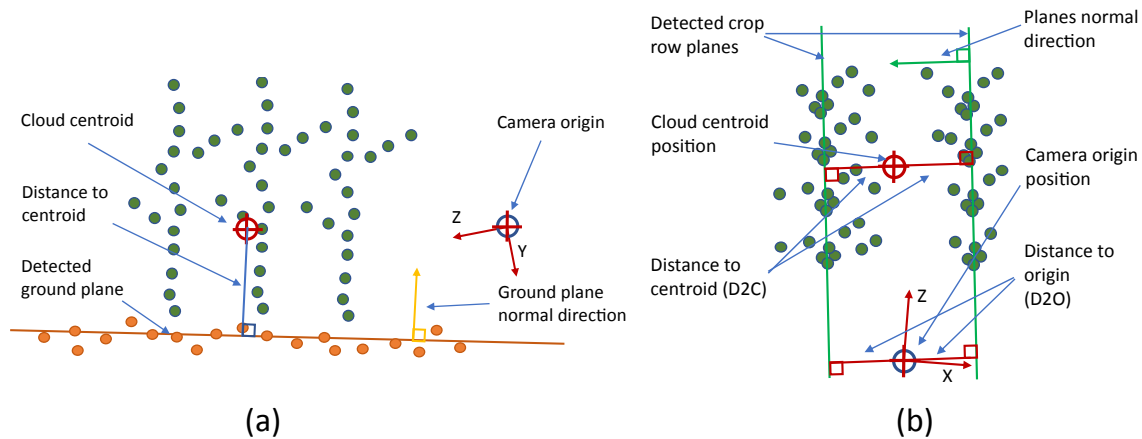


Fig. 9. The illustration of the plane parameters definition. Left: The illustration of the ground plane normal direction and its distance to the cloud centroid from side view. Right: The illustration of the crop row planes normal direction and their distances to the cloud centroid (D2C) and to the camera origin (D2O) from top view.

centroid of each frame was calculated from the point cloud after pre-processing, which remained consistent in experiments and in the ground truth extraction. The accuracy of the ground plane and crop row detection steps were evaluated by the mean absolute error (MAE) of these parameters. The accuracies of crop rows normal directions and D2Os of crop rows were critical for the inter-row positioning task, and the accuracy of D2Cs was critical for the field mapping task.

Fig. 10 shows two example sets of step results of the detection algorithm for both the 30 DAP and 80 DAP datasets. The statistical analysis is presented in Table 1. For the 30 DAP dataset and 80 DAT dataset, 114 out of 120 images (95%) and 144 out of 200 images (72%) were valid for processing, respectively. The invalid images were those with either the ground or the crop rows fully or mostly occluded by leaves of crop plants and weeds. For the valid images, the accuracies of both ground plane estimation and crop row plane estimation were evaluated (Figs. 11–12). For the 30 DAP dataset, the algorithm was able to detect the ground plane with mean absolute error (MAE) of 0.013 rad (0.74 deg) in normal direction, and 0.16 cm in distance to the cloud centroid. And the crop rows were detected with MAE of 0.057 rad (3.2 deg) in normal direction, and 2.2 cm and 6.9 cm in D2C and D2O, respectively.

For the 80 DAP dataset, the algorithm was able to detect the ground plane with MAE of 0.79 rad (4.5 deg) in normal direction, and 2.1 cm in distance to the cloud centroid. The crop rows were detected with MAE of 0.054 rad (3.1 deg) in normal direction, and 3.6 cm and 8.0 cm in D2C and D2O, respectively. The processing time for each frame was about 0.3 s on average, using a computer with a i5-6500TE processor.

Most of the invalid cases observed were caused by the sensor obstruction by the leaves, in which the structural features of crop rows were not perceivable. And the rest of the failures were caused by the tall weeds which affected both the ground detection and row detection. However, we should expect normal weed control practice in research plots, thus the chance of having large weed plants is small.

As observed, performance differences exist in both ground detection and row detection steps on the two different datasets. For the ground detection step, lower accuracy was observed on the 80 DAP dataset, which was mainly caused by the higher weed density and the higher soil moisture level. The weeds affected the RANSAC based ground detection as outliers in the algorithm, and the wet soil absorbed more IR light for ToF measurement, which reduced the measurement range and accuracy of the ToF camera.

For the row detection step, lower accuracy was also observed on the 80 DAP dataset. Cases with obvious errors were caused by two main reasons:

The first is the sensor measurement noise of the soil pixels. The proposed algorithm relies on the stalk pixels, particularly the section close to the ground. The inaccurate ground plane led to difficulties in

extracting these stalk pixels.

The second reason is weed infestation and more complex structure, especially the dropping leaves of the crop plants at late growth stages. They both introduced outliers in the crop row fitting step.

In general, the moderate row detection accuracy in terms of row direction and distance to the cloud centroid (D2C) indicates its high fidelity in field mapping applications. However, the error level in estimating the distance to the camera (D2O) is high, which means lower fidelity in lateral positioning. That was due to the sensor's limitation of a narrow field of view and relatively short measurement range. In this study, the cloud centroid is about 1.5 m away from the camera, which means minor error in row direction would lead to high D2O error.

Comparing to the LiDAR-based algorithm developed by Higuti et al. (2019), the advantage of our proposed algorithm is the ability to detect parallel crop rows with a certain row spacing. However, one limitation is that the row detection results were produced by using only one image, which is prone to outliers such as weeds and leaves, due to the least-square based strategy and relying on no historical measurements. Additionally, the proposed algorithm is computationally more costly because of the higher data density of the ToF camera compared to LiDAR sensors.

3.2. Mapping and lateral positioning performance evaluation

The performance of field mapping and lateral positioning based on the proposed row detection algorithm were evaluated. Field tests in both corn and sorghum fields were conducted at Iowa State University Agricultural Engineering and Agronomy Research Farm located in Boone, Iowa in October of 2019. The plants were planted in late season (August), with a standard 0.76 m (30") row spacing, and about 0.15 m (6") inter-plant spacing. During field tests, the plants were at about the V5 growth stage (about 0.6 m tall, 40 DAP). The soil was dry, and the weather was sunny (60k lux sunlight) with a strong southwest wind (38 km/h, 24 mph). The developed PhenoBot 3.0 was manually driven in the crop rows with a speed of about 1 m/s (~2 mph). The mapping performance was evaluated by comparing the generated map with the manually surveyed crop rows, and the lateral positioning performance was evaluated by comparing the system-derived results with the GPS-IMU fusion localization results.

Fig. 13 shows example run trails for testing field mapping in corn and sorghum fields. Both example run trials are 35 m in length. Among the ToF camera frames during the run trails, 325 out of 504 and 359 out of 401 frames were valid and processed for the corn field trail and the sorghum field trail. As observed, the constructed maps using the proposed algorithm matches the manually surveyed ground truth. The means of the absolute position errors (MAE) were 3.4 cm (1.3") and 3.6

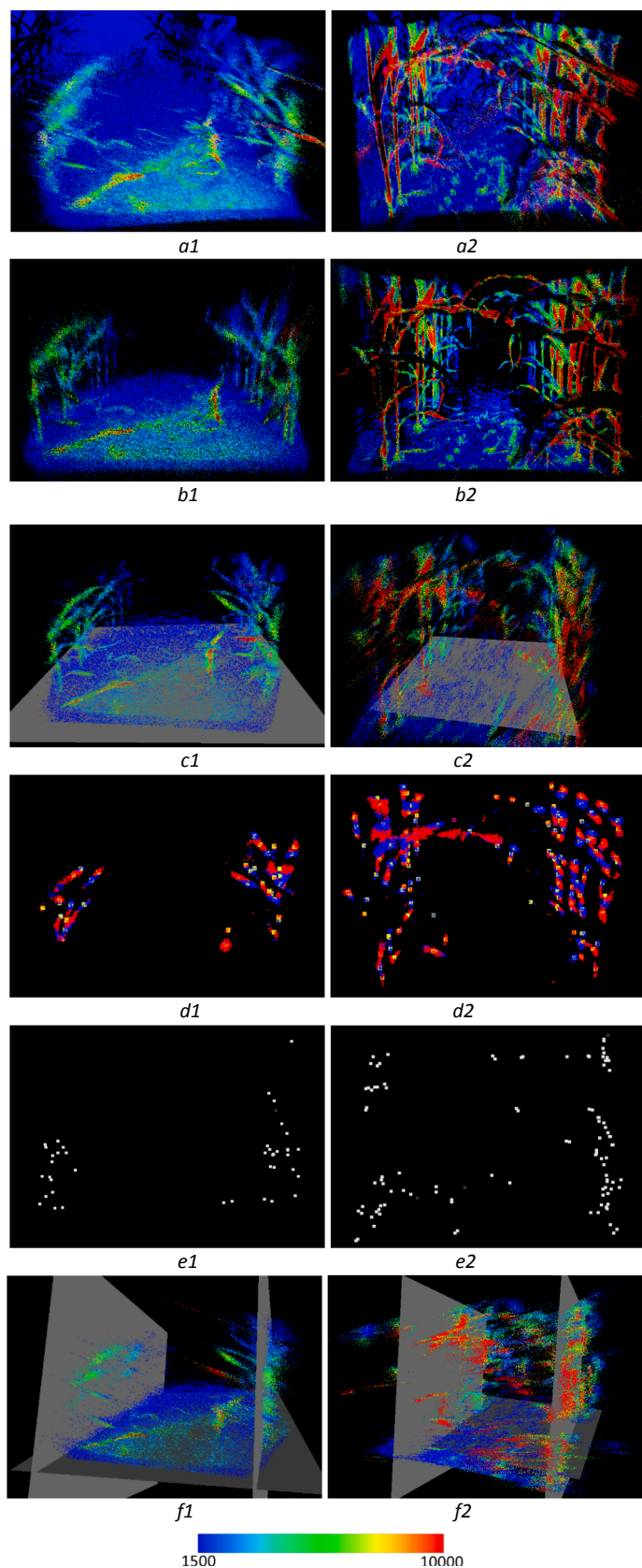


Fig. 10. Sample images of representative step results. (data collected at 30 DAT for left images, 80 DAT for right images) including a) the raw 3D point clouds without pre-processing; b) the 3D point clouds after pre-processing; c) the detected ground planes; d) the extracted vegetation pixels above the ground planes and the clustering results; e) supervoxels projected to top-view, and f) the detected parallel planes representing the crop rows. The points in a), b), c) and f) were colored by the infrared intensity value from the raw images.

Table 1
Crop row detection performance.

Dataset		30 DAP	80 DAP
Valid images/total images		114/120 (95%)	144/200 (72%)
Ground detection accuracy (MAE)	Direction error	0.013 rad (0.74 deg)	0.079 rad (4.5 deg)
	Distance to cloud centroid (D2C) error	1.6×10^{-3} m	0.021 m
Row detection accuracy (MAE)	Direction error	0.057 rad (3.2 deg)	0.054 rad (3.1 deg)
	Distance to cloud centroid (D2C) error	0.022 m	0.036 m
	Distance to camera origin (D2O) error	0.069 m	0.080 m

MAE: mean absolute error compared with the ground truth. DAP: days after planting.

cm (1.4") in corn and sorghum field tests, respectively (Fig. 14). The results are acceptable considering the stalk size of the crop plants, and the requirement of our navigation system. And the results are generally consistent with the algorithm performance evaluation results in the previous section.

Fig. 15 shows example run trials for testing inter-row positioning in corn field and sorghum field. The lateral position and direction of the robot were calculated using images acquired in the mapping tests. As observed, the inter-row positioning results without filtering have a moderate level of error in heading and lateral position. The means of the

absolute heading errors were 0.054 rad (3.0 deg) and 0.055 rad (3.1 deg) in corn and sorghum field tests, respectively (Fig. 16). And the means of the absolute positioning errors were 5.0 cm and 4.2 cm in corn and sorghum field tests, respectively (Fig. 17). The positioning error levels are slightly higher than what presented in the algorithm performance evaluation in the previous section, which was due to the strong south wind during data collection. The leaves were blown to the north, which made the detected row on the south of the robot closer than the ground truth.

As a summary, the results of this study showed that the proposed single ToF-camera-based system was capable of detecting crop rows and contributed to both field mapping and the inter-row positioning for navigation in the field. Compared to the LiDAR-based systems, there are some benefits of using ToF-camera-based systems:

- The critical information was more likely to be perceived as ToF sensors offer a wider vertical field of view compared with the single-layer measurement of LiDAR sensors.
- ToF camera provides 3D point cloud data, in which the soil surface could be extracted as a reference plane for crop row detection.

Although some limitations of LiDAR sensors were overcome, clear limitations were found in using ToF sensors for field-based navigation between crop rows:

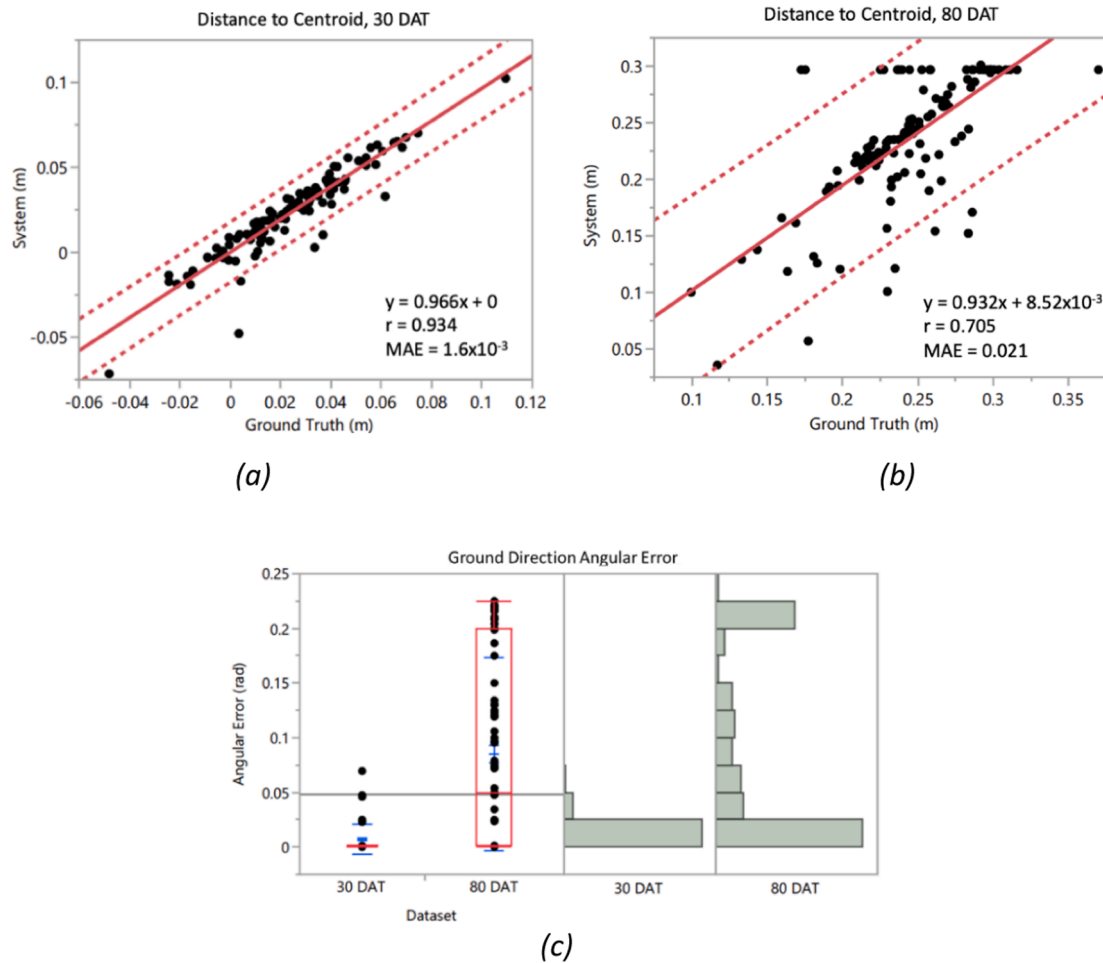


Fig. 11. The ground detection performance on the 30 DAT and 80 DAT datasets. (a, b): The linear regression results of the measured distance to the cloud centroid and the ground truth in both datasets. Strong correlations exist in both datasets. (c): The distribution of the absolute angular error for ground plane normal measurement.

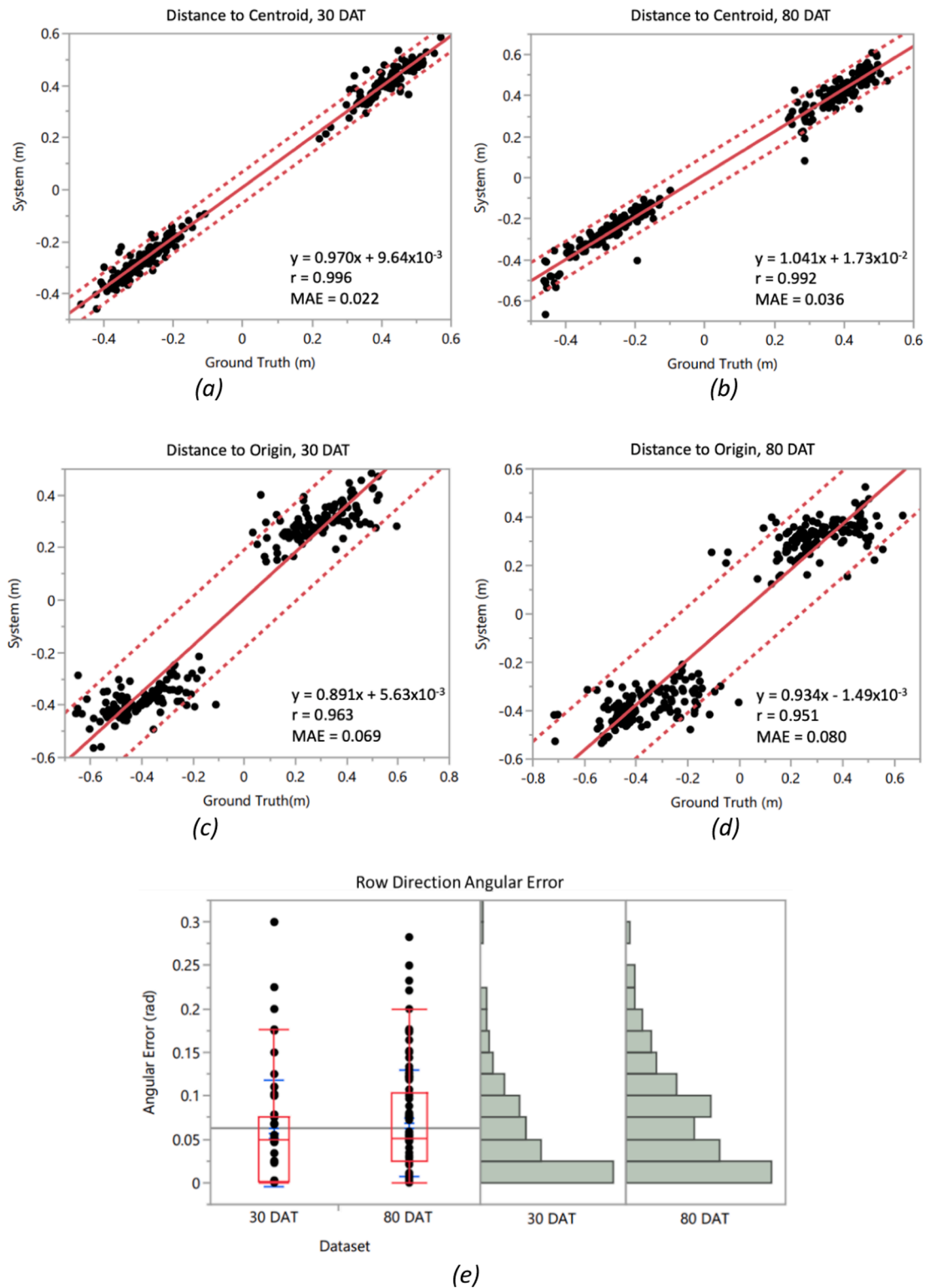


Fig. 12. Row detection performance on the 30 DAT and 80 DAT datasets. (a, b): The linear regression results of the measured distance to the cloud centroid (D2C) and the ground truth in both datasets. (c, d): The linear regression results for measured distance to the camera position (D2O) and the ground truth in both datasets. Strong correlations exist in both D2C and D2O correlation in both datasets. (e): The distribution of the absolute angular error for row direction measurement.

- The narrow horizontal field of view and short measurement range of the ToF camera led to higher uncertainties in inter-row positioning, since a minor error in row direction estimation would produce a large error in lateral position estimation of the robot.
- The data quality of the ToF sensors is affected by the field conditions. For example, strong sunlight and high soil moisture would increase

the measurement noise and reduce the reliable measurement range, therefore reduce the row detection accuracy.

In order to improve the reliability of ToF-camera-based navigation systems, some methods are suggested: a) expand the field of view by introducing other sensors, e.g. another back facing ToF camera on the

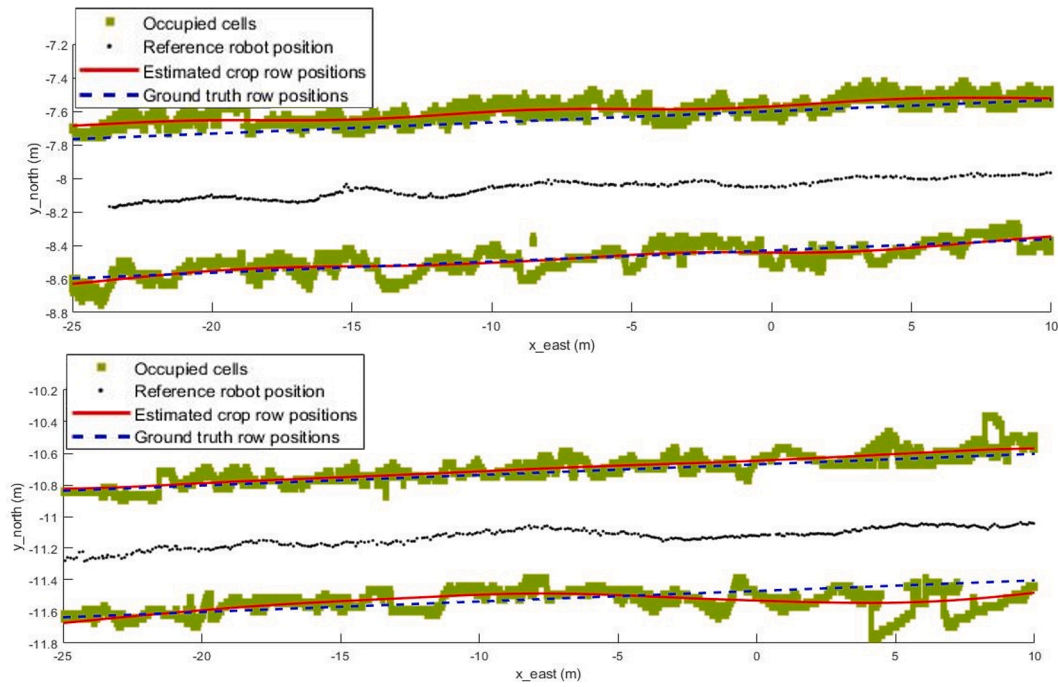


Fig. 13. The mapping results of the crop rows and the ground truth positions of the example field tests of corn (top) and sorghum (bottom).

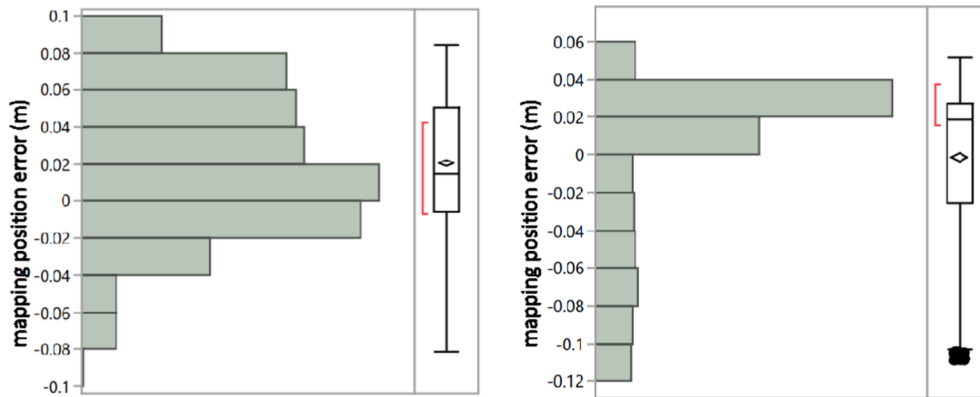


Fig. 14. The distribution of the mapping position error for the corn (left) and sorghum (right) field tests. Unit: m.

robot, or laser range sensors on the side of the robot; b) improve the data reliability by using an auxiliary sensor such as a color camera, which would help extracting critical structural features in row detection; d) fuse the historical data to increase the data density while adding the temporal dimension to the data, which can potentially improve crop row detection, as what proposed in [Higuti et al. \(2019\)](#)'s LiDAR-based system.

4. Conclusions

In this paper, we investigated the feasibility of using a front-facing ToF camera for vision-based navigation and mapping in corn/sorghum fields under crop canopies. In the system, a novel algorithm was developed to detect parallel crop rows from under-canopy depth images using linear programming techniques. Two critical tasks in navigation were accomplished based on the detection results: 1) generating crop field maps as occupancy grids; and 2) positioning the robot in the crop rows. The results of algorithm evaluation showed that our proposed row detection algorithm was able to provide reliable crop row detection results as long as three requirements were met: known crop row spacing, straight crop rows, and flat and level terrain. The proposed system was

evaluated in field tests. The test results showed that the proposed system was able to map the crop rows with mean absolute errors (MAE) of 3.4 cm and 3.6 cm in corn and sorghum fields, respectively. And it provides inter-row positioning data with MAE of 5.0 cm and 4.2 cm for positioning in corn and sorghum crop rows, respectively.

Using a front-facing ToF camera for navigation demonstrated certain feasibilities. The depth information of a ToF camera enables the crop row detection under crop canopies, therefore contribute to field mapping and inter-row positioning for robot navigation. The ToF-camera-based system encounters less sensor occlusion and is less prone to weeds or leaves compared with LiDAR-based systems. However, due to the limited sensing range, narrow horizontal field of view and noisy data quality of the ToF camera, the performance of the inter-row localization is still improvable when using a single sensor data frame each time. Actions that will potentially improve the performance are to fuse the ToF sensor with other sensors, as well as to fuse past historical frames to improve the row detection accuracy.

In summary, the presented algorithm and field-testing results demonstrate the feasibility of using a ToF camera for robotic navigation between crop rows while in the meantime revealed its limitations which can lead to further investigations. The framework of the proposed

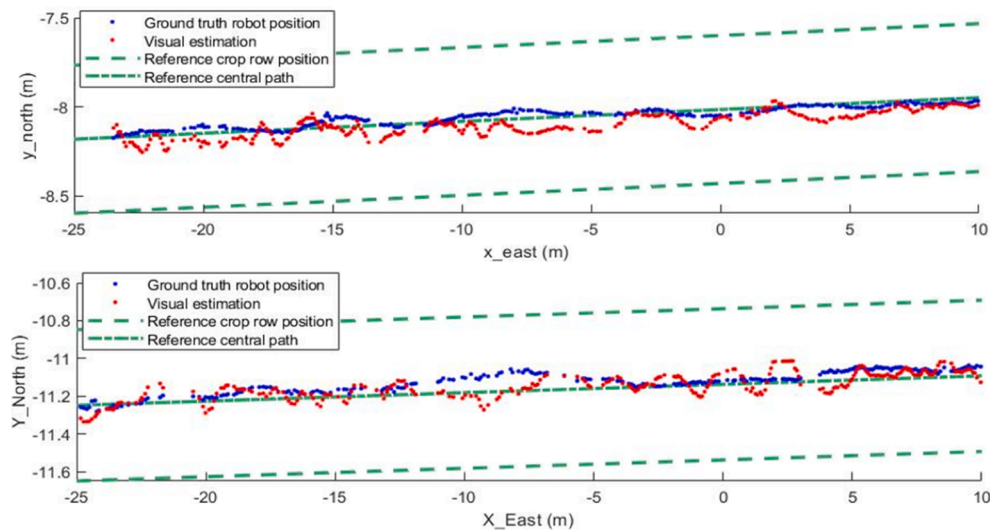


Fig. 15. The visual estimated lateral positioning results and the robot position ground truth of the example field tests of corn (top) and sorghum (bottom).

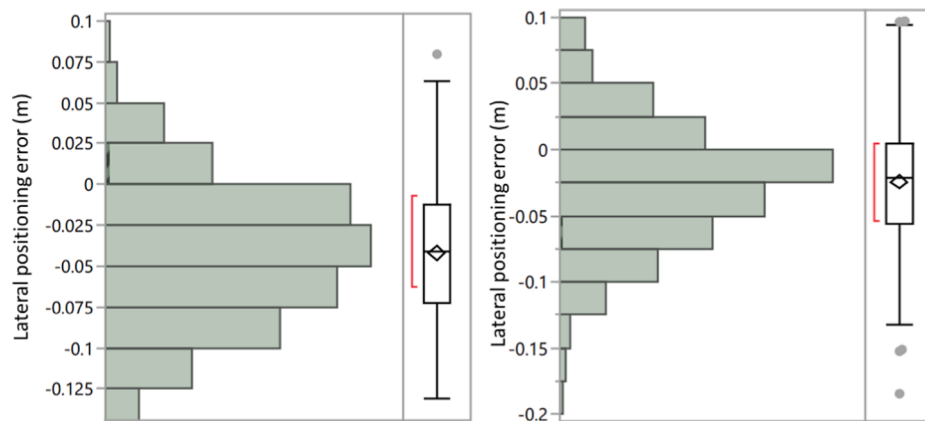


Fig. 16. The distribution of the inter-row lateral positioning error for the corn (left) and sorghum (right) field tests. Unit: m.

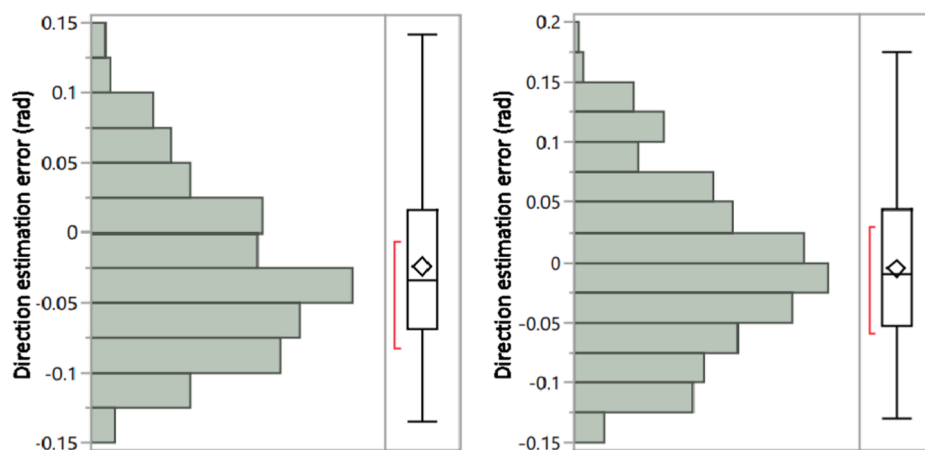


Fig. 17. The distribution of the inter-row direction estimation error for the corn (left) and sorghum (right) field tests. Unit: rad.

algorithm can also be extended to different types of mobile robotic operations under crop canopies.

CRedit authorship contribution statement

Jingyao Gai: Methodology, Software, Writing - original draft.

Lirong Xiang: Software, Data curation, Writing - review & editing. Lie Tang: Conceptualization, Project administration, Funding acquisition, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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